

The Ephemeris

September 2022

Volume 33 Number 03- The Official Publication of the San Jose Astronomical Association



September 2022 - December 2022 Events

Schedule subject to change due to Covid-19 restrictions.

Please check our website for up to date information.

Board & General Meetings

Saturday 9/10, 10/15, 11/12, 12/10

Board Meetings: 6 -7:30pm

General Meetings: 8-9:30pm

Fix-It Day (2-4pm)

Currently on hold due to COVID-19 Situation

Solar Sunday (2-4pm)

Sunday 9/13

Intro to the Night Sky Class

In-Town Star Party(9-11pm)

Friday 9/23, , 10/14, 11/18, 12/16

RODO Starry Nights Star Party

Saturday 9/17, 10/22, 11/19, 12/17

Imaging SIG Mtg

Tuesday 9/20, 10/18, 11/15, 12/20

Astro Imaging Workshops

(at Little Uvas OSP)

Currently on hold due to COVID-19 Situation

Please refer to the SJAA Website home page

(www.sjaa.net) and click on any "Upcoming Event" high-

lighted in Blue. It will take you to the SJ-Astronomy

Meetup page (www.meetup.com/SJ-Astronomy/

[events/](#)) where you can find specific event times, loca-

tions, maps and possible cancellation due to weather.

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A Dragon in Cygnus

Photo by SJAA Member: Glenn Newell

June 23, 2022

Location: Telescope Live

Glenn C. Newell Copyright 2022

Welcome to SJAA



Introducing Steven Nelson—SJAA Director² and QuickStart Program Leader

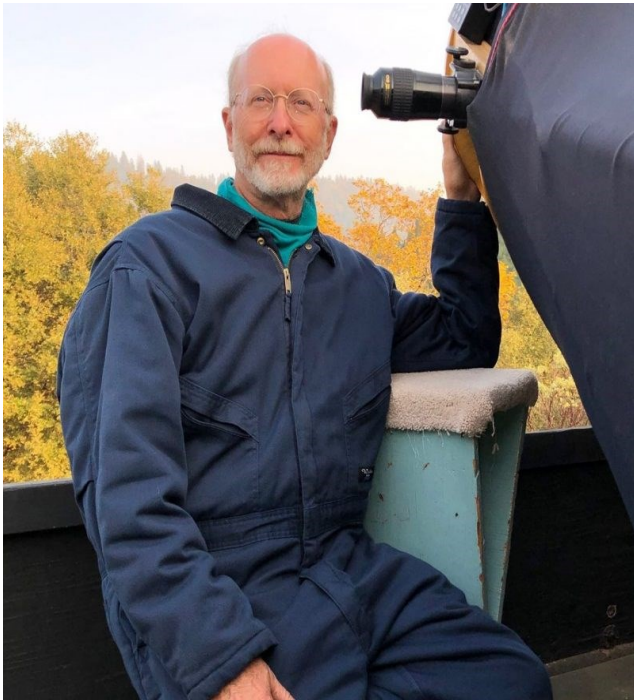
Like our new Donations lead Rich, I majored in Astronomy at UC Berkeley. And in common with Starry Nights lead Joe, I worked on the Hubble's Faint Object Spectrograph (a bit at UC San Diego) and I first joined SJAA around the turn of the century. In my first stint I was also on the Board, when we were really struggling how to be 'a corporation' and have our very first contract (Houge Park with the City of SJ). We did not figure out also how to 'get dark sky land' and fund an observatory.

As a second career STEM teacher (Physical Science) I'm also excited to be more active rejoining the SJAA's public charity mission for public education. I loved working in the lab - and on "hands-on-science" education.

My second hat / QuickSTART lead. I took over this position from the long-time Leadership Team and recent ultra-volunteer Dave Ittner. I have 'lots' of optical telescope experience, both amateur and professional. I recently decided to do a 'telescope CV' to document my 'scope experience. Started grinding/ polishing / figuring my own 6" mirror in 1967-68. 12" reflector refurbish in high school with two buddies ... using 24" photometric reflector at Lick Observatory (college thesis project), and 120" Lick and 88" Mauna Kea observing runs with a Berkeley professor (they were the biggest on the mountain' back in '72). I was the lead commissioning engineer on a 'digital' 24" RC reflector (Nixie tube, TTL era!) in London, and later joined astronomical instrumentation groups at UC San Diego (Mt. Lemmon AZ 60") and UC Santa Cruz/Lick Observatory.

For about a decade my "main scope" was a 10" Meade LX-200 SCT (GoTo) and in the last decade of retirement a fine (heavy) 16" 'transportable' Dob. Like Joe (again) my new baby is an eVscope!

In Memoriam Don Machholz (1952-2022)



Former San Jose resident, SJAA Member, comet hunter, and one of the inventors of the annual Messier Marathon, Don Machholz passed away yesterday at age 69. The announcement came via an online post from Don's wife, "It is with profound sadness that I share with everyone that my beloved husband, Don Machholz, passed away unexpectedly and swiftly from COVID at 3:15 AM on Tuesday, August 9, 2022."

Don did a weekly podcast, "Looking Up With Don". The final episode was posted on Monday, August 8 - <https://donmachholz.com/podcast/luwd-136-saturn-at-opposition/>

During part of that episode, Don recounts one of his comet discoveries on the morning of August 13, 1994. At 11:40 of the podcast he talks about calling friends from SJAA to let them know about the new comet since there was an SJAA star party scheduled that evening.

Don also contributed articles to the EarthSky Website which posted this article - <https://earthsky.org/human-world/don-machholz-1952-2022/>

Don will be missed in the astronomy community. Rest in peace Don.

Joe Fragola

Don is legendary. And kind. After The Ephemeris recently ran an article thanking Jim Van Nuland for his donations, Don took the time to write to Jim at his care facility. The letter was warm and personal, very heart-felt. I had been looking through back issues of our newsletter a few months ago, and came across numerous of his contributions, and recognized him from EarthSky articles and his comments. I didn't realize he had a podcast too! I am saddened by his death and extend my sympathy to his wife and family.

Marianne Damon

Don was an active local member when I first joined SJAA around 'the turn of the century'. In my opinion he exemplified the type of member - not an official 'elected leader' - who vastly contributed to our club in many ways. He wrote a monthly - Comet Comments* Ephemeris section starting before the internet / World Wide Web was widely available (i.e. IAU circulars) with comet information. He explained what was going on with these Shallow Sky objects. He gave us some talks, and some articles, on his 'productive' sky scanning styles (and places - like the backside of Mt. Umunhum). After he 'left us' for the darker sky of the Sierra foothills, he continued his California public education efforts.

Steven Nelson

Origin of the Constellations

Credit: Joe Fragola

Most of the constellation names we know came from the ancient Middle Eastern, Greek, and Roman cultures. They identified groups of stars as gods, goddesses, animals, and objects of their folklore. But these were not the only cultures filling the night sky with characters important to their lives. Cultures all over the world, such as Native American, Asian, and African, have made pictures with the same stars. In some cases the constellations may have had ceremonial or religious significance. In other cases, the star groupings helped to mark the passage of time between planting and harvesting. There are 50 “ancient” constellations viewable with the unaided eye from the Northern Hemisphere.

Additional modern constellations such as Microscopium the Microscope, Telescopium the telescope, and Horologium the clock were identified by later astronomers of the 1500s, 1600s, and 1700s who used telescopes and who were able to observe the night sky in the southern hemisphere. These scientists connected the dimmer stars between the ancient constellations. There are 38 of these modern constellations. In 1930 the International Astronomical Union officially listed 88 modern and ancient constellations and established the boundary around each one.

So that’s how we got to where we are today. But what about going back to when constellations were originally created by the various cultures mentioned above? In addition to vivid imaginations, they also had the fortune of observing the night sky without having to deal with the scourge of light pollution. Even with the advantage of pristine skies, the outlines of some constellations seemed incomplete. The Native American Navajo tribe had a story to explain why this is so. Here is part of their star lore handed down through generations.

A long, long, time ago there was only enough sun to last for half the time, and so in half a day the sun must move across the sky.

And in those first of days, each evening when the sun lay down beyond the western hills and waters, the sky above stood dark and empty.

The people said, “We cannot find our way along the paths. We are afraid.”

So, the Great Spirit who created the world called a great council of the animals to help him. And to the creatures who assembled at the council meeting he said, “Go to the river and gather up the tiny sparkling stones.”

He took one tiny rock himself and placed it in the sky, saying, “This stone will be called a “star” and it will be the “home star”, a campfire that will never move. Look for it if you are lost. It will help you find your way. All the other stars will revolve around it.”

Then he said to the animals, “Now each of you take some sparkling rocks, as many as you can carry, and draw a picture of yourself in the sky.” Each of the creatures set to work. But most of the animals were too small to carry enough stones to complete their pictures. So the Great Spirit gave Coyote a large bag of sparkling stones so that Coyote could help the smaller creatures. But Coyote grew impatient. He took the stones and flung them into the sky, which is why some of the star pictures are unfinished and why the stars don’t all form clear patterns. It was only then that Coyote realized he had forgotten his own picture, and now there were no rocks left. So Coyote howled, and still today a coyote howls at the night sky because his picture is not there.

It is interesting to see how folklore such as this is used to explain nature. It is also interesting that there are some star groupings which represent the same animal, object, or person across different cultures, even though these cultures were separated by thousands of miles. For folks interested in exploring star patterns from another perspective, the free planetarium software Stellarium allows you to change from the traditional constellation overlays to others based on different cultures.

Are there SJAA Members who know some fun and interesting star lore from their cultures? I’d be interested in hearing these other stories and encourage you to share them with others.

Comet Headed Our Way

Credit: Joe Fragola

Here's an early look at a comet that the astronomy community will probably be talking about as we get closer to the New Year. Comet C/2022 E3 ZTF was discovered on March 2, 2022. ZTF stands for Zwicky Transient Facility, a robotic survey camera installed on the 48-inch Samuel Oschin Schmidt Telescope at the Palomar Observatory in southern California. The ZTF scans the entire Northern sky every two days looking for "transient" sky phenomena such as comets, asteroids, and supernovae.

Currently located in Hercules, this 12th magnitude comet is heading for the inner solar system. Closest approach to the Sun will be on January 13, 2023 at a distance of 1.11 AU. The comet will be closest to Earth on February 2, 2023 when it passes 0.29 AU (27 million miles) from us. In February Comet ZTF will be conveniently placed high in the northern sky in the faint constellation of Camelopardalis, the Giraffe. It could reach the limit of naked eye visibility, and will certainly be easily visible in binoculars with a predicted peak magnitude of 5.5. Barring discovery of a new brighter comet during the coming months, or unexpected break-up of the comet before perihelion, I expect the media will pick up on this comet and you'll be hearing more about it as we get closer to the end of the year.

The image attached below was collected on August 18 at Rancho Canada del Oro (5-minute integration time). I showed Comet ZTF as one of my targets during last night's In Town Star Party at Houge Park. I plan to show it again tonight at the Starry Nights public star party at RCDO. Hopefully this comet follows predicted brightening and we'll get a nice view to close out 2022 and start the New Year.



Comet C/2022 E3 ZTF

Credit: Joe Fragola

Webb Captures Stellar Gymnastics in The Cartwheel Galaxy

Credit: NASA



NASA's James Webb Space Telescope has peered into the chaos of the Cartwheel Galaxy, revealing new details about star formation and the galaxy's central black hole. Webb's powerful infrared gaze produced this detailed image of the Cartwheel and two smaller companion galaxies against a backdrop of many other galaxies. This image provides a new view of how the Cartwheel Galaxy has changed over billions of years.

The Cartwheel Galaxy, located about 500 million light-years away in the Sculptor constellation, is a rare sight. Its appearance, much like that of the wheel of a wagon, is the result of an intense event – a high-speed collision between a large spiral galaxy and a smaller galaxy not visible in this image. Collisions of galactic proportions cause a cascade of different, smaller events between the galaxies involved; the Cartwheel is no exception.

A large pink, speckled galaxy resembling a wheel with a small, inner oval, with dusty blue in between on the right, with two smaller spiral galaxies about the same size to the left against a black background.

Credits: NASA, ESA, CSA, STScI

The collision most notably affected the galaxy's shape and structure. The Cartwheel Galaxy sports two rings — a bright inner ring and a surrounding, colorful ring. These two rings expand outwards from the center of the collision, like ripples in a pond after a stone is tossed into it. Because of these distinctive features, astronomers call this a "ring galaxy," a structure less common than spiral galaxies like our Milky Way.

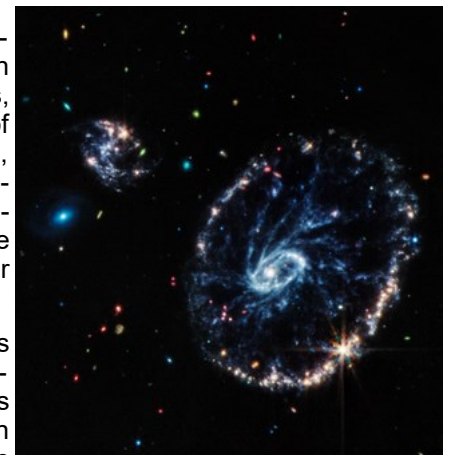
The bright core contains a tremendous amount of hot dust with the brightest areas being the home to gigantic young star clusters. On the other hand, the outer ring, which has expanded for about 440 million years, is dominated by star formation and supernovas. As this ring expands, it plows into surrounding gas and triggers star formation.

Other telescopes, including the Hubble Space Telescope, have previously examined the Cartwheel. But the dramatic galaxy has been shrouded in mystery – perhaps literally, given the amount of dust that obscures the view. Webb, with its ability to detect infrared light, now uncovers new insights into the nature of the Cartwheel.

The Near-Infrared Camera (NIRCam), Webb's primary imager, looks in the near-infrared range from 0.6 to 5 microns, seeing crucial wavelengths of light that can reveal even more stars than observed in visible light. This is because young stars, many of which are forming in the outer ring, are less obscured by the presence of dust when observed in infrared light. In this image, NIRCam data are colored blue, orange, and yellow. The galaxy displays many individual blue dots, which are individual stars or pockets of star formation. NIRCam also reveals the difference between the smooth distribution or shape of the older star populations and dense dust in the core compared to the clumpy shapes associated with the younger star populations outside of it.

Learning finer details about the dust that inhabits the galaxy, however, requires Webb's Mid-Infrared Instrument (MIRI). MIRI data are colored red in this composite image. It reveals regions within the Cartwheel Galaxy rich in hydrocarbons and other chemical compounds, as well as silicate dust, like much of the dust on Earth. These regions form a series of spiraling spokes that essentially form the galaxy's skeleton. These spokes are evident in previous Hubble observations released in 2018, but they become much more prominent in this Webb image.

Webb's observations underscore that the Cartwheel is in a very transitory stage. The galaxy, which was presumably a normal spiral galaxy like the Milky Way before its collision, will continue to transform. While Webb gives us a snapshot of the current state of the Cartwheel, it also provides insight into what happened to this galaxy in the past and how it will evolve in the future.



This image from Webb's Mid-Infrared Instrument (MIRI) shows a group of galaxies, including a large distorted ring-shaped galaxy known as the Cartwheel

Credits: NASA, ESA, CSA, STScI, Webb ERO Production Team

SJAA Member Astrophoto Gallery

WR134, featuring "The Eye in the Sky"

Francesco Meschia



© Francesco Meschia

Dates: Aug. 8, 2022- Aug. 14, 2022, Aug. 17, 2022, Aug. 19, 2022

Location: Mountain View, CA

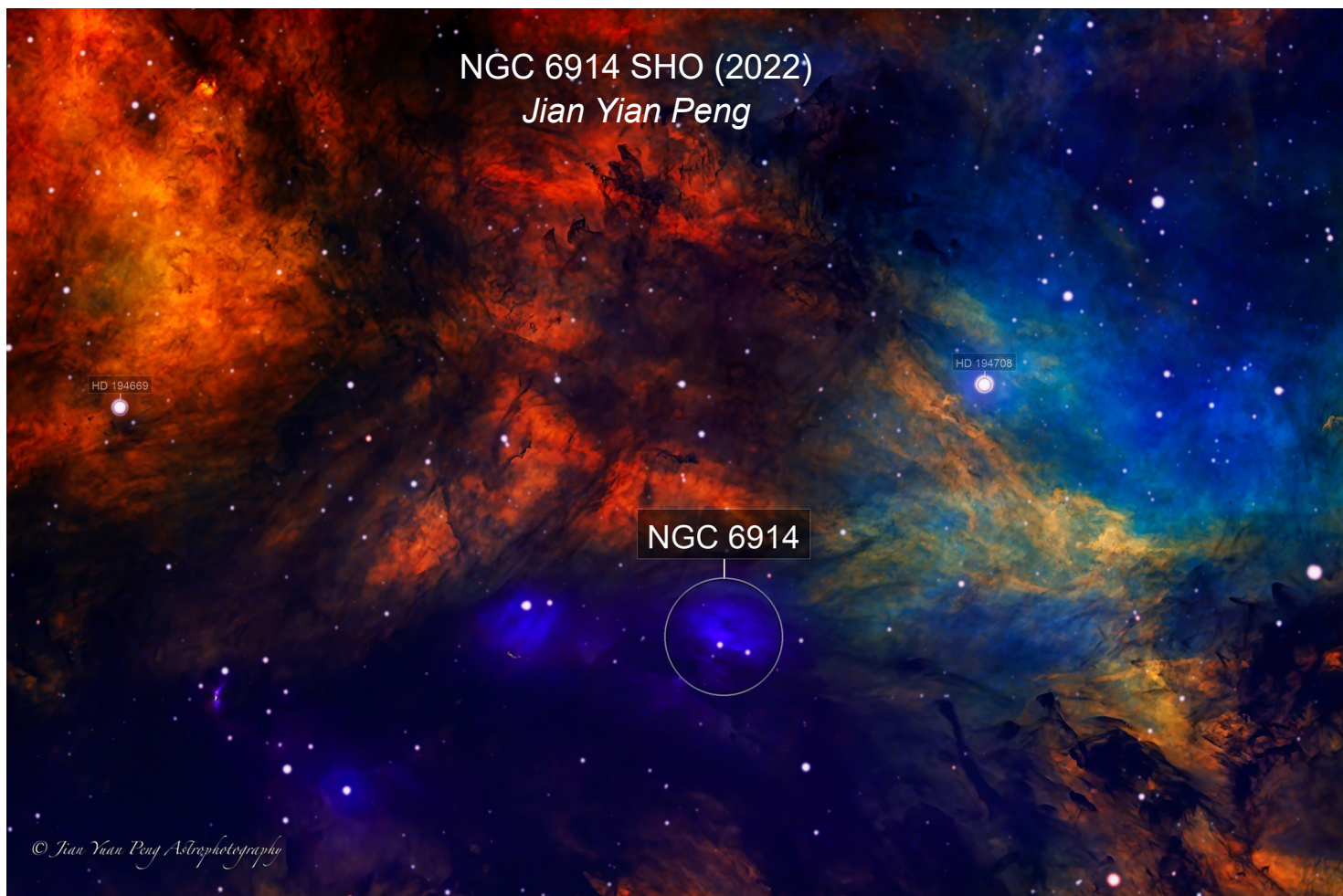
Imaging Telescope: Astro-Tech AT130EDT

Imaging Camera: ZWO ASI294MM Pro

Mount: iOptron iEQ45 Pro

Software: Pleiades Astrophoto PixInsight , Russell Croman Astrophotography NoiseXTerminator , Russell Croman Astrophotography StarXTerminator , Stefan Berg Nighttime Imaging 'N' Astronomy (N.I.N.A./NINA)

NGC 6914 SHO (2022)
Jian Yian Peng



Date: Aug 10, 2022

Location: Palo Alto, CA

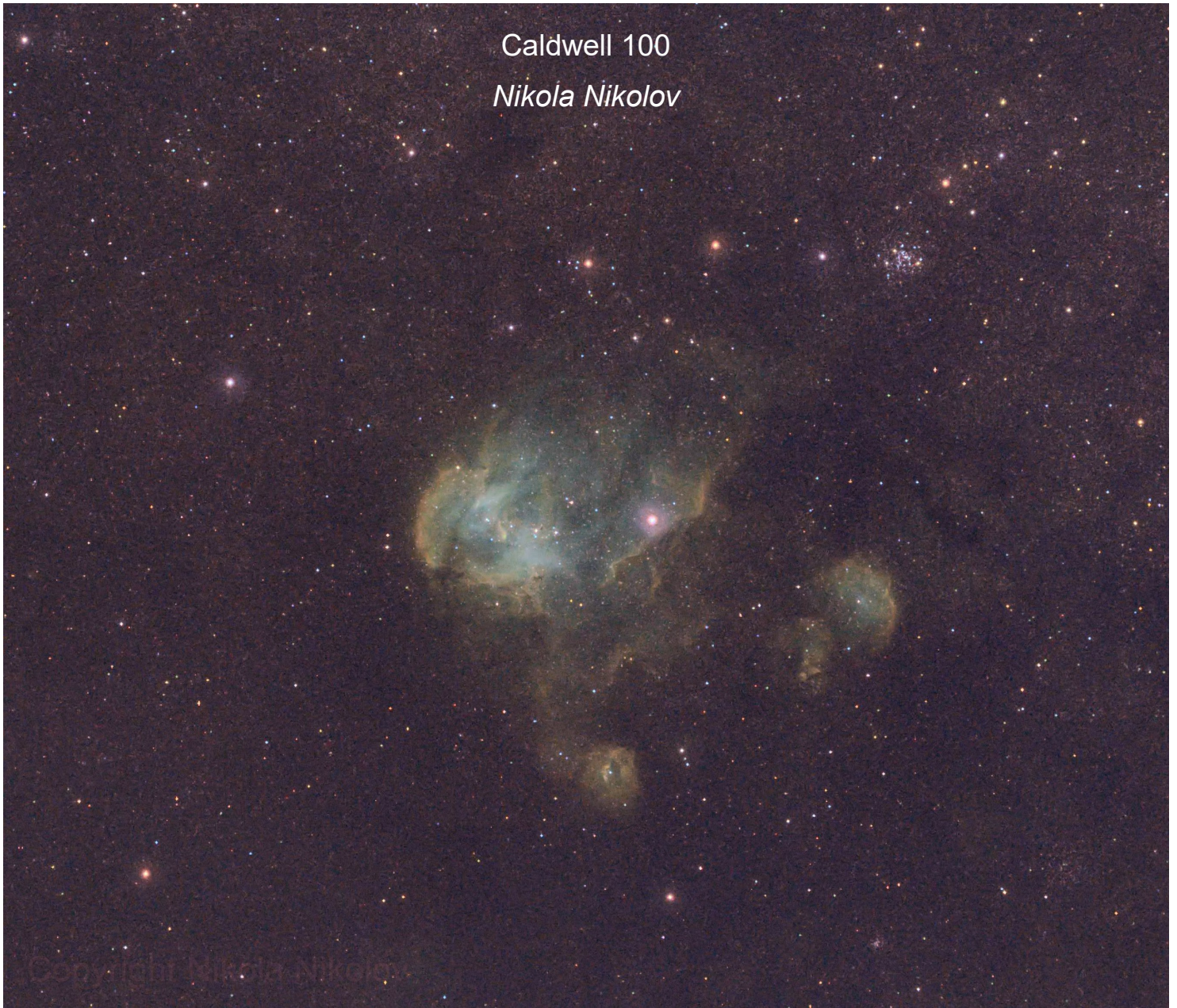
Telescope: Celestron Edge HD11 + Reducer 0.7x

Camera: ZWO ASI2600MM Pro

Mount: AP Mach2GTO

Software: PHD2, PixInsight, Lightroom Classic, Sequence Generator Pro, Photoshop

Caldwell 100
Nikola Nikolov



Dates: 5 August 2022

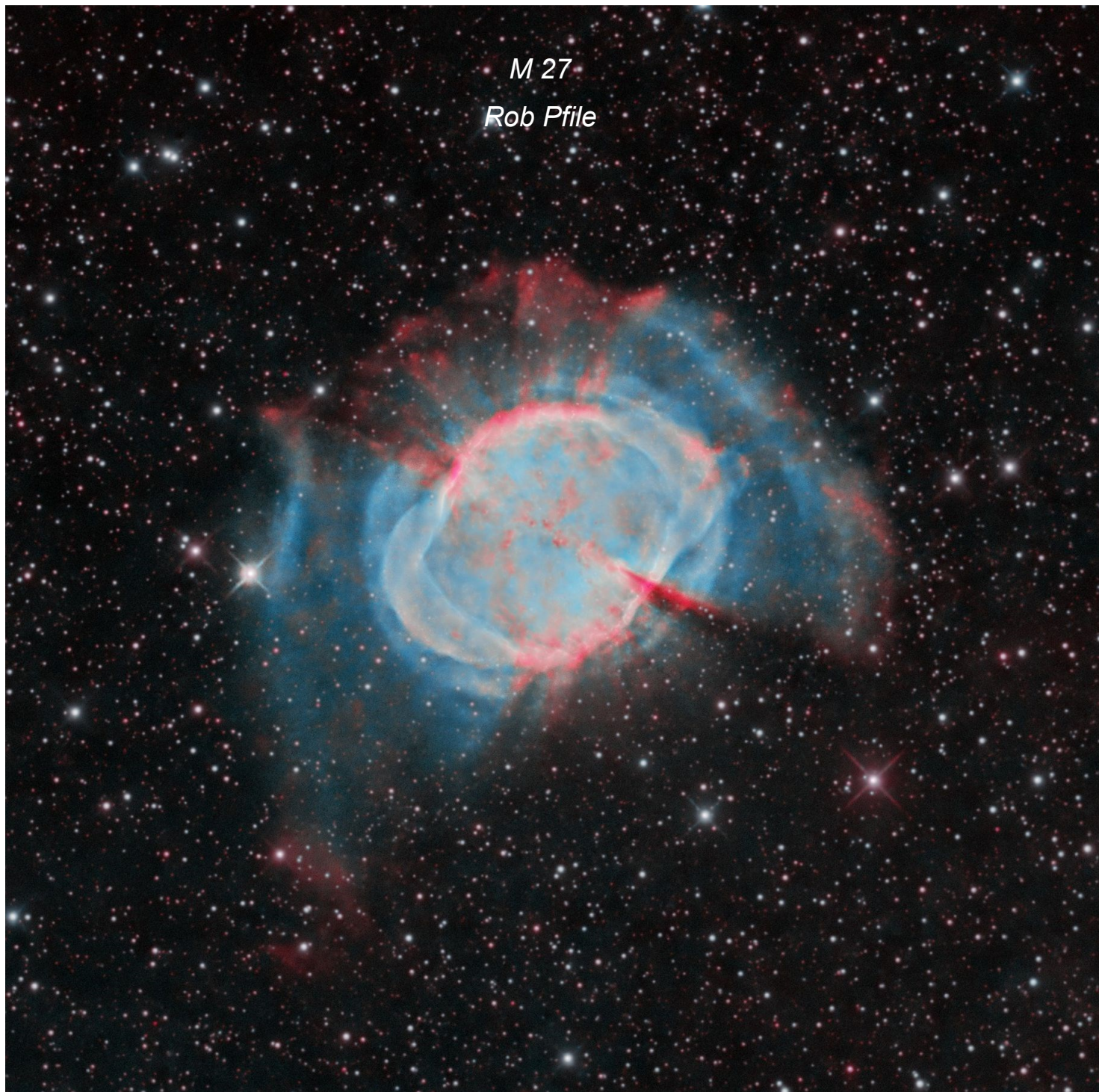
Location: El Sauce Observatory, Rio Hurtado Valley, Coquimbo Region, Chile

Telescope: Samyang 135mm F2.0 ED UMC

Camera: ZWO ASI1600MM

Mount: Rainbow Astro RST-135

Software: Adobe Lightroom, Aries Productions Astro Pixel Processor (APP), Pleiades Astrophoto PixInsight, Silicon Fields StarTools



M 27
Rob Pfile

Date: July 1, 2022

Location: Casa de Rob, Oakland, CA

Telescope: Astro-Tech AT6RC

Camera: ZWO ASI2600MM Pro

Mount: Astro-Physics Mach1 GTO

Software: Pleiades Astrophoto PixInsight , Russell Croman Astrophotography NoiseXTerminator , Russell Croman Astrophotography StarXTerminator , Stefan Berg Nighttime Imaging 'N' Astronomy (N.I.N.A. / NINA)

Using Cepheids As Distance Measures

Credit: Jordan Makower

There are several ways to measure the distance to stars. This method demonstrates one way, with a certain class of variable stars, Cepheids. These stars expand and contract over a period of time, related to their mass. This results in changes to the star's brightness. The difference between the star's brightest and dimmest appearance (apparent magnitude) and the length of time over which this happens is used to determine its true brightness (absolute Magnitude). The observer fills out the chart below after careful observations, then uses additional charts/graphs to determine the star's distance.

The star, Delta, in the constellation Cepheus, is used for this exercise. This constellation changes position in the Northern sky during several hours. Most of the stars seem to form the outline of a house with a tall, peaked roof. We will concentrate on the stars at the bottom left labeled Delta (δ), Zeta (ζ), and Epsilon (ϵ). Zeta is the brightest of the three, Epsilon is dimmest, but Delta varies in brightness.

You should observe these stars whenever they are visible, recording the date&time. You should also record Delta's relative brightness compared to the other two stars, by filling out this chart (which refers to Delta's brightness) with an **X**.

When you have enough accurate, honest data, you will be able to determine the period of time that this star takes to vary from bright to dim to bright again. You will use that information to compute its distance from Earth.

Making your observations: If you look at Delta, Zeta, and Epsilon, and you see 2 bright stars and 1 dim star, DELTA is at its brightest. If you look at those stars and you see 1 bright one and 2 dim ones, DELTA is at its dimmest. You may need many days of observations until you can determine Delta's variation period, so the chart will have to be extended, and may take up a few sheets of paper.

Once you are sure of Delta's variation period, use the next chart (from Burnham's Celestial Handbook, Vol. 1) to determine Delta's true brightness (absolute Magnitude). The difference between this number and its average apparent magnitude can be used in a formula (Distance Modulus) to calculate its distance.

Astronomers use the magnitude numbering system in this way; the lower the number, the brighter the object. Very bright objects, such as the Moon = - 12.6, our Sun = - 26.7, the star Sirius = - 1.4, Vega = 0.0.

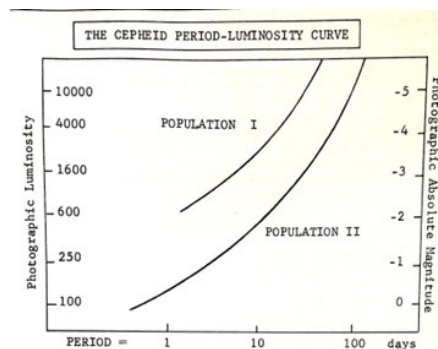
When Delta is as bright as Zeta, its apparent magnitude is about 3.4. When Delta is as dim as Epsilon, its apparent magnitude is about 4.2. Its average apparent magnitude, then, is about 3.8. Scientists calculate a star's true brightness by mathematically 'viewing' its brightness at a distance of 32.6 light years. Our Sun's true brightness is only 4.83 ! Why does it seem so bright to us.

If you want to mathematically determine Delta's distance from us, here is an excerpt from *A Dictionary of Astronomy*:

The difference between the apparent magnitude of an object, m , corrected for interstellar absorption, and its absolute magnitude, M . This difference is directly linked to the distance in parsecs, r , by the formula: $m - M = 5 \log (r/10)$.

Put another way, if the distance modulus is known, the distance in parsecs can be found from $r = 10 \times 1.585(m - M)$. (The part in parentheses should be shown in superscript as a 'power')

Delta's	Date	
Bright		
Not-so-bright		
In-between		
Not-so-dim		
Dim		



THE PERIOD-LUMINOSITY RELATION OF CEPHEIDS FOR POPULATION I, FROM THE ABOVE GRAPH			
PERIOD = Abs.Pg.Mag.		PERIOD = Abs.Pg.Mag.	
1d	-1.5	13d	-3.6
2	-2.1	14	-3.7
3	-2.4	15	-3.8
4	-2.7	16	-3.9
5	-2.9	18	-4.0
6	-3.0	20	-4.1
7	-3.2	25	-4.3
8	-3.3	30	-4.5
9	-3.4	35	-4.6
10	-3.5	40	-4.8
11	-3.5	45	-4.9
12	-3.6	50	-5.0



Hubble Views a Galactic Marvel

Credit: NASA



Credit: ESA/Hubble & NASA, R. B. Tully, R. Jansen, R. Windhorst

The galaxy featured in this image from NASA's Hubble Space Telescope has a shape unlike many of the galaxies familiar to Hubble. Its thousands of bright stars evoke a spiral galaxy, but it lacks the characteristic 'winding' structure. The shining red blossoms stand out as well, twisted by clouds of dust – these are the locations of intense star formation. The galaxy also radiates a diffuse glow, much like an elliptical galaxy and its core of older, redder stars. This galactic marvel is known to astronomers as NGC 1156.

NGC 1156 is located around 25 million light-years from Earth, in the constellation Aries. It has a variety of different features that are of interest to astronomers. A dwarf irregular galaxy, it's also classified as isolated, meaning no other galaxies are nearby enough to influence its odd shape and continuing star formation. The extreme energy of freshly formed young stars gives color to the galaxy, against the red glow of ionized hydrogen gas, while its center is densely packed with older generations of stars.

Hubble has captured NGC 1156 before. This new image features data from a galactic gap-filling program simply titled "Every Known Nearby Galaxy." Astronomers noticed that Hubble had observed only three quarters of the galaxies within just over 30 million light-years of Earth in sufficient detail to study the makeup of the stars within them. They proposed that in between larger projects, Hubble could take snapshots of the remaining quarter, including NGC 1156. Gap-filling programs like this ensure the best use of Hubble's valuable observing time.

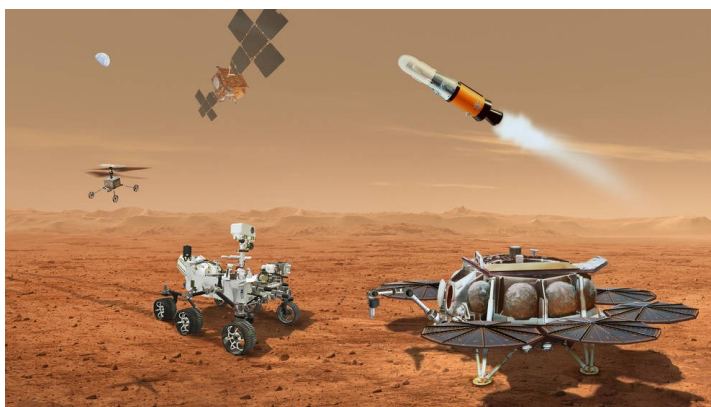


Kid Spot Jokes:

- ♦ **A photon walks into a hotel and the clerk asks if he needs help with luggage.**
- ♦ "No thanks!" says the photon. "I'm traveling light."
- ♦ **I was teaching my son astronomy last night and he chuckled when I pointed out the dog star.**
- ♦ Hey, I said this is Sirius!

NASA Will Inspire World When It Returns Mars Samples to Earth in 2033

Credit: NASA



This illustration shows a concept for multiple robots that would team up to ferry to Earth samples collected from the Mars surface by NASA's Mars Perseverance rover.

Credits: NASA/JPL-Caltech

NASA has finished the system requirements review for its Mars Sample Return Program, which is nearing completion of the conceptual design phase. During this phase, the program team evaluated and refined the architecture to return the scientifically selected samples, which are currently in the collection process by NASA's Perseverance rover in the Red Planet's Jezero Crater.

The architecture for the campaign, which includes contributions from the European Space Agency (ESA), is expected to reduce the complexity of future missions and increase probability of success.

"The conceptual design phase is when every facet of a mission plan gets put under a microscope," said Thomas Zurbuchen, associate administrator for science at NASA Headquarters in Washington. "There are some significant and advantageous changes to the plan, which can be directly attributed to Perseverance's recent successes at Jezero and the amazing performance of our Mars helicopter."

This advanced mission architecture takes into consideration a recently updated analysis of Perseverance's expected longevity. Perseverance will be the primary means of transporting samples to NASA's Sample Retrieval Lander carrying the Mars Ascent Vehicle and ESA's Sample Transfer Arm.

As such, the Mars Sample Return campaign will no longer include the Sample Fetch Rover or its associated second lander. The Sample Retrieval Lander will include two sample recovery helicopters, based on the design of the Ingenuity helicopter, which has performed 29 flights at Mars and survived over a year beyond its original planned lifetime. The helicopters will provide a secondary capability to retrieve samples cached on the surface of Mars.

The ESA Earth Return Orbiter and its NASA-provided Capture, Containment, and Return System remain vital elements of the program architecture.

With planned launch dates for the Earth Return Orbiter and Sample Retrieval Lander in fall 2027 and summer 2028, respectively, the samples are expected to arrive on Earth in 2033.

With its architecture solidified during this conceptual design phase, the program is expected to move into its preliminary design phase this October. In this phase, expected to last about 12 months, the program will complete technology development and create engineering prototypes of the major mission components.

This refined concept for the Mars Sample Return campaign was presented to the delegates from the 22 participating states of Europe's space exploration program, Terrae Novae, in May. At their next meeting in September, the states will consider the discontinuation of the development of the Sample Fetch Rover.

From the Board of Directors

Announcements

- ◇ . San Jose Astronomical Association is closely monitoring the corona virus COVID-19 situation in Santa Clara County. Since the mask mandate restriction was lifted as of June 3, SJAA has gradually reconvened the in-person activities, such as In Town Star Party, Starry Nights Star Party, and Solar Observation.
- ◇ Please check our Meetup to determine if an event has been moved online.

Board Meeting Excerpts

June 18, 2022

In attendance

Rob Jaworski, Vini Carter, Peter Melhus, Ken Miura, Steven Nelson, Wolf Witt, Glen Newell, Aroon Nair, Swami Nigam

Absences: Sukhada Palav

Guests: Joe, Kanch

HP building readiness for SJAA operation

The SJAA board planned to discuss certain concerns with the Shakespeare Group, regarding stuff lying around the SJAA section of the Houge Park location.

Fiscal Year change

Rob J called for moving the fiscal year to be July 1 - June 30. Peter seconded. Steven Nelson voted against. All others voted yes.

Sign-in sheets

The board discussed and agreed with the use of sign-in sheets for all events at Houge Park.

July 16, 2022

In attendance

Peter Melhus, Ken Miura, Steven Nelson, Aroon Nair, Swami Nigam, Vini Carter

Absences: Rob Jaworski, Sukhada Palav, Wolf Witt, Glen Newell

Guests: Kanch, Nancy Lieu, Joe Fragola, Jon Anderson, Leslie Carter

Code of Conduct for SJAA leadership Team

Steven moved to approve the changes to a document that is to be signed by everyone who holds a position at SJAA. The name, position, signature, and date have to be mentioned in the document. Peter seconded the motion. Ken called for a vote. The motion was carried.

Protection of Minors Policy

As per City of San Jose attorneys, SJAA is to have a documented protection of minors policy. Swami moved to approve a simple screening process done for volunteers at the club's expense. Peter seconded, and the motion was approved unanimously.

Funds for Solar Program

Ken moved to approve the requested amount of \$500 to purchase anti-glow filters for the club's Lunt H-alpha telescopes, that have a background glow making it difficult for kids to view images satisfactorily. Steven seconded the motion, and it was voted for unanimously.

August 13, 2022

In attendance

Vini Carter, Steven Nelson, Ken Miura, Swami Nigam, Sukhada Palav, Peter Melhus, Glen Newell, Aroon Nair, Rob Jaworski

Absent: Wolf Witt

Guests: Carl Wong, Joe Fragola, Renuka Veeraswamy

Maintenance of bylaw document for the club

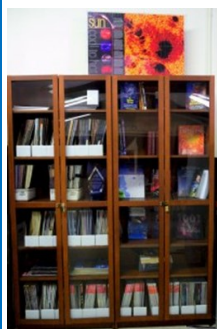
There have been several updates/amendments to the bylaws. SJAA will be working on having the bylaw document ready for approval by the Board in September.

New Compliance: annual reporting requires asset inventory

SJAA plans to have program leaders to assess and provide the value of the inventory assets in their respective programs to comply with ByLaws and

State laws stating charities are required to have a reasonable accounting of "equipment" assets in the Annual Report.

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Steven Nelson, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick START program is now taking signups from members. In general will operate around ITSP in very small groups outside.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will

go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Hy Murveit the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

<http://www.sjaa.net/programs/school-star-party/>

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SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

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Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

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